1.43 Matrices: Determinants

Worked Solutions

1)

$|\mathbf{A}| = ad - bc$

2 a)

$$\begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 3 \times 2 - 1 \times 4 = 6 - 4 = 2$$

2 b)

$$\begin{vmatrix} 6 & -4 \\ 3 & 5 \end{vmatrix} = 6 \times 5 - (-4) \times 3 = 30 + 12 = 42$$

2 c)

$$\begin{vmatrix} 3 & 2 \\ 8 & -5 \end{vmatrix} = 3 \times (-5) - 2 \times 8 = -15 - 16 = -31$$

2 d)

$$\begin{vmatrix} -6 & -5 \\ 3 & 2 \end{vmatrix} = -6 \times 2 - (-5) \times 3 = -12 + 15 = 3$$

3 a)

$$\begin{vmatrix} k & 5 \\ 2k & 3 \end{vmatrix} = k \times 3 - 5 \times 2k = 3k - 10k = -7k$$

3 b)

$$\begin{vmatrix} k & 4 \\ 5 & 3k \end{vmatrix} = k \times 3k - 4 \times 5 = 3k^2 - 20$$

3 c)

$$\begin{vmatrix} 3k & k+1 \\ 4 & k-2 \end{vmatrix} = 3k \times (k-2) - (k+1) \times 4 = 3k(k-2) - 4(k+1)$$
$$= 3k^2 - 6k - 4k - 4 = 3k^2 - 10k - 4$$

3 d)

$$\begin{vmatrix} k-2 & k-3 \\ k & 2k+1 \end{vmatrix} = (k-2)(2k+1) - (k-3)k$$
$$= 2k^2 - 3k - 2 - k^2 + 3k = k^2 - 2$$

4 a)

$$\begin{vmatrix} p & q \\ q & p \end{vmatrix} = p \times p - q \times q = p^2 - q^2$$

4 b)

$$\begin{vmatrix} p & p-3 \\ q & q+2 \end{vmatrix} = p(q+2) - (p-3)q = pq + 2p - pq + 3q = 2p + 3q$$

4 c)

$$\begin{vmatrix} q-1 & p-1 \\ p+1 & q+1 \end{vmatrix} = (q-1)(q+1) - (p-1)(p+1) = (q^2-1) - (p^2-1) = q^2 - p^2$$

4 d)

$$\begin{vmatrix} q+2 & p+3 \\ p-4 & q-3 \end{vmatrix} = (q+2)(q-3) - (p+3)(p-4)$$
$$= (q^2 - q - 6) - (p^2 - p - 12) = q^2 - q - p^2 + p + 6$$

5)

$$\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = \cos\theta \times \cos\theta - (-\sin\theta) \times \sin\theta = \cos^2\theta + \sin^2\theta = 1$$

6 a)

$$\begin{vmatrix} \ln 8 & \ln 2 \\ \ln 4 & \ln 16 \end{vmatrix} = \begin{vmatrix} \ln 2^3 & \ln 2 \\ \ln 2^2 & \ln 2^4 \end{vmatrix} = \begin{vmatrix} 3\ln 2 & \ln 2 \\ 2\ln 2 & 4\ln 2 \end{vmatrix}$$
$$= (3\ln 2)(4\ln 2) - (\ln 2)(2\ln 2) = 12(\ln 2)^2 - 2(\ln 2)^2 = 10(\ln 2)^2$$

6 b)

$$\begin{vmatrix} \ln 81 & \ln 27 \\ \ln 9 & \ln 3 \end{vmatrix} = \begin{vmatrix} \ln 3^4 & \ln 3^3 \\ \ln 3^2 & \ln 3 \end{vmatrix} = \begin{vmatrix} 4\ln 3 & 3\ln 3 \\ 2\ln 3 & \ln 3 \end{vmatrix}$$
$$= (4\ln 3)(\ln 3) - (3\ln 3)(2\ln 3) = 4(\ln 3)^2 - 6(\ln 3)^2 = -2(\ln 3)^2$$

7 a)

Area $= \frac{1}{2} \times 4 \times 2 = 4$ units^2.

7 b)

$\det \mathbf{A} = 2 \times 5 - 1 \times 3 = 10 - 3 = 7$

7 c)

$$\begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 0 & 4 & 2 \\ 0 & 0 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 8 & 2 \\ 0 & 12 & -4 \end{pmatrix}$$

$$\text{Area} = 16 \times 8 - \frac{1}{2} \times 8 \times 12 - \frac{1}{2} \times 2 \times 4 - \frac{1}{2} \times 6 \times 16 = 28 \text{ units}^2$$

7 d)

The determinant is the area scale factor, i.e. 4 units$^2 \times 7 = 28$ units^2.

8)

The area of Q is $6 \times 15 = 90$ units^2.

9)

The smallest possible area of D is $3150 \times 24.5 = 77175$ units^2.

10)

$$\begin{vmatrix} 5 & 8 \\ 2 & 6 \end{vmatrix} = 5 \times 6 - 8 \times 2 = 30 - 16 = 14$$

5 cm$^2 \times 14 = 70$ cm^2. So the area of S' is 70 cm^2.

11 a)

Area $= 8 \times 14 - \frac{1}{2} \times 4 \times 6 - \frac{1}{2} \times 8 \times 8 - \frac{1}{2} \times 4 \times 14 = 40$ units^2.

11 b)

$\det \mathbf{A} = 3 \times 6 - (-2) \times 4 = 18 + 8 = 26$

11 c)

40 units$^2 \times 26 = 1040$ units^2.

12 a)

$$\begin{pmatrix} 5-k & 3 \\ -4 & 4 \end{pmatrix} \begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix} = \begin{pmatrix} 19-2k & 5-4k & k-8 \\ 4 & -36 & 0 \end{pmatrix}$$

So the coordinates of the vertices of T' are $(19-2k, 4)$, $(5-4k, -36)$ and $(k-8, 0)$.

12 b)

$$\det \mathbf{A} = (5-k) \times 4 - 3 \times (-4) = 4(5-k) + 12 = 20 - 4k + 12 = 32 - 4k$$

12 c)

Area of $T = 5 \times 8 - \frac{1}{2} \times 3 \times 4 - \frac{1}{2} \times 5 \times 4 - \frac{1}{2} \times 2 \times 8 = 16$ units^2.

$$16 \text{ units}^2 \times (32 - 4k) = 320 \text{ units}^2$$

$$16(32 - 4k) = 320 \Rightarrow 32 - 4k = 20 \Rightarrow 4k = 12 \Rightarrow k = 3$$

13 a)

$$\begin{pmatrix} 4-k & 2 \\ -6 & k+3 \end{pmatrix} \begin{pmatrix} -4 & 1 & 2 \\ 1 & 6 & -7 \end{pmatrix} = \begin{pmatrix} 4k-14 & 16-k & -2k-6 \\ k+27 & 6k+12 & -7k-33 \end{pmatrix}$$

So the coordinates of the vertices of T' are $(4k-14, k+27)$, $(16-k, 6k+12)$ and $(-2k-6, -7k-33)$.

13 b)

$$\det \mathbf{A} = (4-k)(k+3) - 2(-6) = (4-k)(k+3) + 12 = -k^2 + k + 12 + 12 = -k^2 + k + 24$$

13 c)

Area of $T = 13 \times 6 - \frac{1}{2} \times 5 \times 5 - \frac{1}{2} \times 6 \times 8 - \frac{1}{2} \times 1 \times 13 = 35$ units^2.

$$35 \text{ units}^2 \times (-k^2 + k + 24) = 140 \text{ units}^2$$

$$35(-k^2 + k + 24) = 140 \Rightarrow -k^2 + k + 24 = 4 \Rightarrow k^2 - k - 20 = 0$$

$$\Rightarrow (k-5)(k+4) = 0 \Rightarrow k = 5 \text{ only as } k \text{ is a positive constant}$$

14)

Area of $S = 7 \times 7 = 49$ units^2.

$$\det \mathbf{A} = (2k+1)k - 5k \times 3 = (2k+1)k - 15k = 2k^2 + k - 15k = 2k^2 - 14k$$

$$49 \text{ units}^2 \times (2k^2 - 14k) = 784 \text{ units}^2$$

$$49(2k^2 - 14k) = 784 \Rightarrow k^2 - 7k = 8 \Rightarrow k^2 - 7k - 8 = 0 \Rightarrow (k-8)(k+1) = 0$$

$\Rightarrow k = -1$ or $k = 8$.

15)

$$\begin{vmatrix} 60 & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{vmatrix} = 60 \times \frac{1}{2\sqrt{2}} - 0 \times 0 = 15\sqrt{2}$$

$$\text{Area of } T = \frac{1}{2} \times 0.025 \times (\sqrt{2} + \sqrt{8}) = \frac{3\sqrt{2}}{80} \text{ cm}^2$$

$$\text{Area of } T' = 15\sqrt{2} \times \frac{3\sqrt{2}}{80} = \frac{9}{8} = 1.125 \text{ cm}^2$$

16 a)

$\det \mathbf{A} = 6 \times 1 - 2 \times 2 = 6 - 4 = 2$. Area of $T' = 8 \times 2 = 16$ units^2.

16 b)

$\det \mathbf{C} = 6 \times 3 - (-2) \times 1 = 18 + 2 = 20$. Area of $T' = 8 \times 20 = 160$ units^2.

16 c)

$\det \mathbf{B} = (-2) \times (-3) - 4 \times 2 = 6 - 8 = -2$. Area of $T' = 8 \times 2 \times |-2| = 32$ units^2.

16 d)

Area of $T' = 8 \times |-2| \times 2 = 32$ units^2.

16 e)

Area of $T' = 8 \times 2 \times 20 = 320$ units^2.

16 f)

Area of $T' = 8 \times |-2| \times 20 = 320$ units^2.

16 g)

Area of $T' = 8 \times 2 \times |-2| \times 20 = 640$ units^2.

17 a)

$k \times 1 - 3 \times 5 = 3 \Rightarrow k - 15 = 3 \Rightarrow k = 18$.

17 b)

$k \times 4 - 8 \times 5 = 16 \Rightarrow 4k - 40 = 16 \Rightarrow 4k = 56 \Rightarrow k = 14$.

17 c)

$k \times k - 3 \times 4 = 4 \Rightarrow k^2 - 12 = 4 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$.

17 d)

$k \times 9 - 5 \times k = 40 \Rightarrow 9k - 5k = 40 \Rightarrow 4k = 40 \Rightarrow k = 10$.

17 e)

$k \times 1 - k \times 3 = -6 \Rightarrow k - 3k = -6 \Rightarrow -2k = -6 \Rightarrow k = 3$.

17 f)

$$k \times 10 - k \times k = 9 \Rightarrow 10k - k^2 = 9 \Rightarrow 0 = k^2 - 10k + 9$$
$$\Rightarrow 0 = (k-9)(k-1) \Rightarrow k = 9 \text{ or } k = 1$$

17 g)

$$2k \times k - k \times 4 = 2 \Rightarrow 2k^2 - 4k = 2 \Rightarrow k^2 - 2k - 1 = 0 \Rightarrow k = 1 \pm \sqrt{2}$$

17 h)

$$3k \times 4k - 6 \times k = 6 \Rightarrow 12k^2 - 6k = 6 \Rightarrow 2k^2 - k - 1 = 0$$
$$\Rightarrow (2k+1)(k-1) = 0 \Rightarrow k = -\frac{1}{2} \text{ or } k = 1$$

17 i)

$$(k+1)k - 3 \times 4 = 8 \Rightarrow k^2 + k - 12 = 8 \Rightarrow k^2 + k - 20 = 0$$
$$\Rightarrow (k+5)(k-4) = 0 \Rightarrow k = -5 \text{ or } k = 4$$

17 j)

$$(2k-1)(k-2) - 4 \times 5 = 7 \Rightarrow 2k^2 - 5k + 2 - 20 = 7 \Rightarrow 2k^2 - 5k - 25 = 0$$
$$\Rightarrow (2k+5)(k-5) = 0 \Rightarrow k = -\frac{5}{2} \text{ or } k = 5$$

17 k)

$$k \times 6 - (k-4)(2k+5) = -6 \Rightarrow 6k - (2k^2 - 3k - 20) = -6$$
$$\Rightarrow 6k - 2k^2 + 3k + 20 = -6 \Rightarrow 0 = 2k^2 - 9k - 26$$
$$\Rightarrow (2k-13)(k+2) = 0 \Rightarrow k = \frac{13}{2} \text{ or } k = -2$$

17 l)

$$3 \times 7k - (5k+1)(8-k) = 15 \Rightarrow 21k - (-5k^2 + 39k + 8) = 15$$
$$\Rightarrow 21k + 5k^2 - 39k - 8 = 15 \Rightarrow 5k^2 - 18k - 23 = 0$$
$$\Rightarrow (5k-23)(k+1) = 0 \Rightarrow k = \frac{23}{5} \text{ or } k = -1$$

18)

$$\det \mathbf{A} > 0 \Rightarrow 2 \times 8 - k \times 3 > 0 \Rightarrow 16 - 3k > 0 \Rightarrow -3k > -16 \Rightarrow k < \frac{16}{3}$$

19)

$$\det \mathbf{A} < 0 \Rightarrow 3k \times (-6) - (-4) \times 2 < 0 \Rightarrow -18k + 8 < 0 \Rightarrow -18k < -8 \Rightarrow k > \frac{4}{9}$$

20)

$$\det \mathbf{A} > 0 \Rightarrow k(4k-4) - (k+2) \times 2 > 0$$
$$\Rightarrow 4k^2 - 4k - (2k+4) > 0 \Rightarrow 4k^2 - 6k - 4 > 0$$
$$\Rightarrow 2k^2 - 3k - 2 > 0 \Rightarrow (2k+1)(k-2) > 0$$
$$\Rightarrow k < -\frac{1}{2} \text{ or } k > 2$$

21)

$$\begin{aligned}\det \mathbf{A} < 0 &\Rightarrow (k-1)(2k+1) - (1-k)k < 0 \\ &\Rightarrow 2k^2 - k - 1 - (k - k^2) < 0 \Rightarrow 2k^2 - k - 1 - k + k^2 < 0 \\ &\Rightarrow 3k^2 - 2k - 1 < 0 \Rightarrow (3k+1)(k-1) < 0 \\ &\Rightarrow -\frac{1}{3} < k < 1\end{aligned}$$

22)

$$\det \mathbf{A} = (2+i)i - 3(-2) = 2i + i^2 + 6 = 2i - 1 + 6 = 5 + 2i$$

23 a)

Rotations maintain orientation, so matrices that represent rotations will have a positive determinant.

23 b)

Reflections reverse orientation, so matrices that represent reflections will have a negative determinant.

23 c)

Enlargements maintain orientation, so matrices that represent enlargements will have a positive determinant.

24 a)

$k = \pm 1$.

24 b)

$k > 0$.

25 a)

$$\begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = 1 \times (-1) - 0 \times 0 = -1 < 0$$

As the determinant is negative, the matrix reverses orientation, and so represents a reflection.

25 b)

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0 \times 0 - 1 \times 1 = -1 < 0$$

As the determinant is negative, the matrix reverses orientation, and so represents a reflection.

25 c)

$$\begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix} = (-1) \times (-1) - 0 \times 0 = 1 > 0$$

As the determinant is positive, the matrix maintains orientation, and so does not represent a reflection.

25 d)

$$\begin{vmatrix} 0 & -1 \\ 0 & 1 \end{vmatrix} = 0 \times 1 - (-1) \times 0 = 0$$

As the determinant is zero, all points are mapped to the origin, and so does not represent a reflection.

25 e)

$$\begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} = (-1) \times 1 - 0 \times 0 = -1 < 0$$

As the determinant is negative, the matrix reverses orientation, and so represents a reflection.

25 f)

$$\begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix} = 0 \times 0 - (-1) \times (-1) = -1 < 0$$

As the determinant is negative, the matrix reverses orientation, and so represents a reflection.

26 a)

$$\begin{vmatrix} 4 & -6 \\ -9 & 3 \end{vmatrix} = 4 \times 3 - (-6) \times (-9) = 12 - 54 = -42 < 0$$

As the determinant is negative, the matrix reverses orientation.

26 b)

$$\begin{vmatrix} -5 & 7 \\ 1 & -2 \end{vmatrix} = (-5) \times (-2) - 7 \times 1 = 10 - 7 = 3 > 0$$

As the determinant is positive, the matrix maintains orientation.

26 c)

$$\begin{vmatrix} 8 & -9 \\ 10 & -3 \end{vmatrix} = 8 \times (-3) - (-9) \times 10 = -24 + 90 = 66 > 0$$

As the determinant is positive, the matrix maintains orientation.

26 d)

$$\begin{vmatrix} 21 & 10 \\ 15 & 7 \end{vmatrix} = 21 \times 7 - 10 \times 15 = 147 - 150 = -3 < 0$$

As the determinant is negative, the matrix reverses orientation.

27 a)

Area $= 10 \times 18 = 180$ units^2.

27 b)

A', D', C', B'.

28)

As $k^2 \geq 0$ for all real values of k, we must have that $4k^2 + 2 \geq 2$ for all real values of k. So $\det A = 4k^2 + 2 > 0$ for all real values of k and hence cannot reverse orientation.

29)

The transformation will reverse orientation if $\det A = 9 - 4k^2 < 0$.

$$\Rightarrow (3 - 2k)(3 + 2k) < 0 \Rightarrow k < -\frac{3}{2} \text{ or } k > \frac{3}{2}$$

i.e. $k \in (-\infty, -\frac{3}{2}) \cup (\frac{3}{2}, \infty)$.

30)

$$|A| = 80 \times 2 - (3k) \times 9 = 160 - 27k$$

The transformation will reverse orientation if $\det A = 160 - 27k < 0$.

$$\Rightarrow k > \frac{160}{27} = 5.9259\ldots$$

As k is a positive integer, the smallest value of k will be 6.

31 a)

$$\det A = 3 \times 1 - 2 \times 2 = 3 - 4 = -1$$

The area of S is 1 unit^2. The area of S' is $|-1| \times 1 = 1$ unit^2.

31 b)

As $\det A = -1 < 0$, the orientation of the vertices has been reversed.

32 a)

The area of the square $ABCD$ is 1, and the area of the rectangle $A'B'C'D'$ is 2. The orientation of the vertices has been reversed. So $\det A = -2$.

32 b)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 6 & 6 & 4 \\ 1 & 1 & 2 & 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a+2b & a+3b & 2a+3b & 2a+2b \\ c+2d & c+3d & 2c+3d & 2c+2d \end{pmatrix} = \begin{pmatrix} 4 & 6 & 6 & 4 \\ 1 & 1 & 2 & 2 \end{pmatrix}$$

$$a + 2b = 4 \quad (1) \qquad a + 3b = 6 \quad (2)$$

$\therefore a = 0$, $b = 2$.

$$c + 2d = 1 \quad (3) \qquad c + 3d = 1 \quad (4)$$

$\therefore c = 1$, $d = 0$. $\therefore A = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$.

33)

$$\det A = \frac{3}{5} \times (-\frac{3}{5}) - \frac{3\sqrt{3}}{5} \times \frac{3\sqrt{3}}{5} = -\frac{9}{25} - \frac{27}{25} = -\frac{36}{25}$$

The scale factor of the enlargement is $\sqrt{\frac{36}{25}} = \frac{6}{5}$.

34 a)

$$\det A = 4\sqrt{3} \times \frac{\sqrt{3}}{2} - (-4) \times \frac{1}{2} = 6 + 2 = 8$$

The scale factor of the stretch is 8.

34 b)

$$\begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 4\sqrt{3} & -4 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \Rightarrow \begin{pmatrix} 8a & 8b \\ c & d \end{pmatrix} = \begin{pmatrix} 4\sqrt{3} & -4 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

$$8a = 4\sqrt{3} \Rightarrow a = \frac{\sqrt{3}}{2} \qquad 8b = -4 \Rightarrow b = -\frac{1}{2}$$

$c = \frac{1}{2}$ and $d = \frac{\sqrt{3}}{2}$. $\therefore \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$ is the rotation matrix.

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

$$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{30°, 330°\} \qquad \sin\theta = \frac{1}{2} \Rightarrow \theta = \{30°, 150°\}$$

So the matrix represents a rotation, anticlockwise about the origin, by 30°.

35 a)

$$\det A = (k-2)(k+2) - (k-3)(k+3) = (k^2 - 4) - (k^2 - 9) = 5 > 0$$

So $\det A > 0$ for all values of k.

35 b)

As $\det A > 0$, the matrix A must preserve orientation.

36)

$$AB = \begin{pmatrix} 8 & -5 \\ k & -2 \end{pmatrix} \begin{pmatrix} 9 & 2 \\ 13 & k \end{pmatrix} = \begin{pmatrix} 7 & 16 - 5k \\ 9k - 26 & 0 \end{pmatrix}$$

$\det AB = -1$ as AB preserves area (so $|\det AB| = 1$) but reverses orientation (must be negative).

$$\begin{aligned} &7 \times 0 - (16 - 5k)(9k - 26) = -1 \\ &\Rightarrow -(-45k^2 + 274k - 416) = -1 \\ &\Rightarrow -45k^2 + 274k - 416 = 1 \Rightarrow 0 = 45k^2 - 274k + 417 \\ &\Rightarrow (45k - 139)(k - 3) = 0 \Rightarrow k = \frac{139}{45} \text{ or } k = 3 \end{aligned}$$

37 a)

$\det A = 2 \times 6 - (-4) \times 3 = 12 + 12 = 24.$

37 b)

$\det B = -1 \times 4 - 2 \times (-8) = -4 + 16 = 12.$

37 c)

$$AB = \begin{pmatrix} 2 & -4 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ -8 & 4 \end{pmatrix} = \begin{pmatrix} 30 & -12 \\ -51 & 30 \end{pmatrix}$$

37 d)

$\det AB = 30 \times 30 - (-12) \times (-51) = 900 - 612 = 288.$

37 e)

$\det AB = \det A \times \det B$ as $288 = 24 \times 12$.

38 a)

$\det AB = 5 \times (-6) = -30.$

38 b)

$\det BA = (-6) \times 5 = -30.$

38 c)

$\det ABA = 5 \times (-6) \times 5 = -150.$

38 d)

$\det BABABA = (-6) \times 5 \times (-6) \times 5 \times (-6) \times 5 = -27000.$

39)

$$\det AB = \det A \times \det B \Rightarrow 20 = \det A \times 5 \Rightarrow \det A = 4$$
$$\det A^5 = (\det A)^5 = 4^5 = 1024$$

40)

$$\det AB = \det I = 1 \Rightarrow \det A \times \det B = 1 \Rightarrow \det A = \frac{1}{\det B}$$

$\det A$ and $\det B$ must be reciprocals of one another.

41)

$\det A = 4 \times 2 - k \times 6 = 8 - 6k.$

$$|A^2| = \det(A^2) = (\det A)^2 = 16 \Rightarrow (8 - 6k)^2 = 16$$
$$\Rightarrow 8 - 6k = \pm 4 \Rightarrow k = \frac{8 \pm 4}{6} \Rightarrow k = 2 \text{ or } k = \frac{2}{3}$$

42)

$\det A = 5 \times k - 2 \times (-4) = 5k + 8.$

$$|A^3| = \det(A^3) = (\det A)^3 = 125 \Rightarrow (5k+8)^3 = 125$$

$\Rightarrow 5k + 8 = 5 \Rightarrow k = -\frac{3}{5}.$

43)

$$\det A = 4i - (3+i) \times 2 = 4i - (6+2i) = -6 + 2i$$
$$\det B = (k+1)i - 3(2-i) = (ki+i) - (6-3i) = -6 + ki + 4i = -6 + (k+4)i$$

$$\begin{aligned}\det AB &= \det A \times \det B = (-6+2i)(-6+(k+4)i) \\ &= 36 - 6(k+4)i - 12i - 2(k+4) \\ &= 36 - 6ki - 24i - 12i - 2k - 8 = 28 - 2k - (6k+36)i\end{aligned}$$

If $\det AB$ has argument $\frac{\pi}{4}$ then $\mathrm{Re}(\det AB) = \mathrm{Im}(\det AB)$ and both the real part and imaginary part must be positive.

$$\therefore 28 - 2k = -6k - 36 \Rightarrow 4k = -64 \Rightarrow k = -16$$

Check: when $k = -16$, $\det AB = 60 + 60i$ which has argument $\frac{\pi}{4}$.

44)

$$\begin{aligned}\det A &= (2x-3)(x-2) - (x-5)(x+6) \\ &= (2x^2 - 7x + 6) - (x^2 + x - 30) = x^2 - 8x + 36 \\ &= (x-4)^2 - 16 + 36 = (x-4)^2 + 20\end{aligned}$$

So the minimum value of $\det A$ is 20. So the minimum area of S' is $30 \times 20 = 600$ cm^2. Alternative Method: Let $y = x^2 - 8x + 36$, then $\frac{dy}{dx} = 2x - 8$. Minimum area occurs when $\frac{dy}{dx} = 0$. $\therefore 2x - 8 = 0 \Rightarrow x = 4$. $y|_{x=4} = 20$. So the minimum value of $\det A$ is 20. So the minimum area of S' is $30 \times 20 = 600$ cm^2.

45)

$$\begin{aligned}\det A &= (2x^2+1)(x^2+3) - (4x-3)(4x+3) \\ &= (2x^4 + 7x^2 + 3) - (16x^2 - 9) = 2x^4 - 9x^2 + 12\end{aligned}$$

Let $y = 2x^4 - 9x^2 + 12$, then $\frac{dy}{dx} = 8x^3 - 18x$. Minimum area occurs when $\frac{dy}{dx} = 0$.

$$\therefore 8x^3 - 18x = 0 \Rightarrow 4x^3 - 9x = 0 \Rightarrow (4x^2 - 9)x = 0 \Rightarrow (2x-3)(2x+3)x = 0$$

$\Rightarrow x = \frac{3}{2}$ or $x = -\frac{3}{2}$ or $x = 0$.

$$\frac{d^2y}{dx^2} = 24x^2 - 18$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-\frac{3}{2}} = 36 > 0 \text{ so } x = -\frac{3}{2} \text{ is a local minimum}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=0} = -18 < 0 \text{ so } x = 0 \text{ is a local maximum}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=\frac{3}{2}} = 36 > 0 \text{ so } x = \frac{3}{2} \text{ is a local minimum}$$

$$y|_{x=-\frac{3}{2}} = \frac{15}{8} \text{ and } y|_{x=\frac{3}{2}} = \frac{15}{8}$$

So the minimum value of $\det A$ is $\frac{15}{8}$. So the minimum area of S' is $16 \times \frac{15}{8} = 30$ cm^2.

46)

$$\begin{aligned}
\det A &= (k+1)(k+2) - (k-3)(k-4) = (k^2+3k+2) - (k^2-7k+12) \\
&= 10k - 10 \\
\det B &= (2k-1)(2k-1) - (3k+1)(1-3k) = (4k^2-4k+1) - (-9k^2+1) \\
&= 13k^2 - 4k \\
\det AB = 0 &\Rightarrow \det A \times \det B = 0 \Rightarrow (10k-10)(13k^2-4k) = 0 \\
&\Rightarrow (k-1)(13k-4)k = 0 \Rightarrow k = 0 \text{ or } k = 1 \text{ or } k = \frac{4}{13}
\end{aligned}$$

47)

$$\begin{aligned}
\det A &= (2k+3)(k-4) - (k+1)(k-2) = (2k^2-5k-12) - (k^2-k-2) \\
&= k^2 - 4k - 10 \\
\det B &= (3k+1)(k+2) - (k-3)(k+4) = (3k^2+7k+2) - (k^2+k-12) \\
&= 2k^2 + 6k + 14 \\
\det A = \det B &\Rightarrow k^2 - 4k - 10 = 2k^2 + 6k + 14 \\
&\Rightarrow 0 = k^2 + 10k + 24 \Rightarrow 0 = (k+6)(k+4) \\
&\Rightarrow k = -6 \text{ or } k = -4
\end{aligned}$$

48 a)

i) $\begin{vmatrix} ka & b \\ kc & d \end{vmatrix} = ka \times d - b \times kc = kad - kbc = k(ad-bc) = k\begin{vmatrix} a & b \\ c & d \end{vmatrix}$. ii) $\begin{vmatrix} a & kb \\ c & kd \end{vmatrix} = a \times kd - kb \times c = kad - kbc = k(ad-bc) = k\begin{vmatrix} a & b \\ c & d \end{vmatrix}$. iii) $\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = ka \times d - kb \times c = kad - kbc = k(ad-bc) = k\begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

iv) $\begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix} = ka \times kd - kb \times kc = k^2ad - k^2bc = k^2(ad-bc) = k^2\begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

48 b)

i) $\begin{vmatrix} 6 & 8 \\ -2 & 5 \end{vmatrix} = \begin{vmatrix} 2\times 3 & 2\times 4 \\ -2 & 5 \end{vmatrix} = 2\begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 2 \times 23 = 46$. ii) $\begin{vmatrix} 3 & 12 \\ -2 & 15 \end{vmatrix} = \begin{vmatrix} 3 & 3\times 4 \\ -2 & 3\times 5 \end{vmatrix} = 3\begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 3 \times 23 = 69$. iii) $\begin{vmatrix} 30 & 4 \\ -20 & 5 \end{vmatrix} = \begin{vmatrix} 10\times 3 & 4 \\ 10\times(-2) & 5 \end{vmatrix} = 10\begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 10 \times 23 = 230$. iv) $\begin{vmatrix} 15 & 20 \\ -20 & 50 \end{vmatrix} = \begin{vmatrix} 5\times 3 & 5\times 4 \\ 10\times(-2) & 10\times 5 \end{vmatrix} = 5 \times 10\begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 5 \times 10 \times 23 = 1150$.

49 a)

Calculator:

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 12$$

By hand:

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 1 \begin{vmatrix} 2 & 0 \\ 0 & 6 \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 \\ 0 & 6 \end{vmatrix} + 0 \begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix}$$
$$= 1(2 \times 6 - 0 \times 0) - 0 + 0 = 1 \times 12 = 12$$

49 b)

Calculator:

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 36$$

By hand:

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 3 \begin{vmatrix} 2 & 0 \\ 0 & 6 \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 \\ 0 & 6 \end{vmatrix} + 0 \begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix}$$
$$= 3(2 \times 6 - 0 \times 0) - 0 + 0 = 3 \times 12 = 36$$

49 c)

Calculator:

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{vmatrix} = 27$$

By hand:

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{vmatrix} = 3 \begin{vmatrix} 2 & 3 \\ 1 & 6 \end{vmatrix} - 0 \begin{vmatrix} 0 & 3 \\ 0 & 6 \end{vmatrix} + 0 \begin{vmatrix} 0 & 2 \\ 0 & 1 \end{vmatrix}$$
$$= 3(2 \times 6 - 3 \times 1) - 0 + 0 = 3 \times 9 = 27$$

49 d)

Calculator:

$$\begin{vmatrix} 3 & 0 & 1 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{vmatrix} = 27$$

By hand:

$$\begin{vmatrix} 3 & 0 & 1 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{vmatrix} = 3 \begin{vmatrix} 2 & 3 \\ 1 & 6 \end{vmatrix} - 0 \begin{vmatrix} 0 & 3 \\ 0 & 6 \end{vmatrix} + 1 \begin{vmatrix} 0 & 2 \\ 0 & 1 \end{vmatrix}$$
$$= 3(2 \times 6 - 3 \times 1) - 0 + 1(0 \times 1 - 2 \times 0) = 3 \times 9 + 1 \times 0 = 27$$

49 e)

Calculator:

$$\begin{vmatrix} 3 & 2 & 1 \\ 2 & 2 & 3 \\ 1 & 1 & 6 \end{vmatrix} = 9$$

By hand:

$$\begin{aligned} \begin{vmatrix} 3 & 2 & 1 \\ 2 & 2 & 3 \\ 1 & 1 & 6 \end{vmatrix} &= 3\begin{vmatrix} 2 & 3 \\ 1 & 6 \end{vmatrix} - 2\begin{vmatrix} 2 & 3 \\ 1 & 6 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix} \\ &= 3(2 \times 6 - 3 \times 1) - 2(2 \times 6 - 3 \times 1) + 1(2 \times 1 - 2 \times 1) \\ &= 3 \times 9 - 2 \times 9 + 1 \times 0 = 9 \end{aligned}$$

50)

Calculator:

$$\begin{vmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 1 & 4 \end{vmatrix} = -4$$

By hand:

$$\begin{aligned} \begin{vmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 1 & 4 \end{vmatrix} &= 1\begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} - 1\begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} + 2\begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} \\ &= 1(1 \times 4 - 1 \times 1) - 1(3 \times 4 - 1 \times 1) + 2(3 \times 1 - 1 \times 1) \\ &= 1 \times 3 - 1 \times 11 + 2 \times 2 = -4 \end{aligned}$$

Volume of $C' = 4 \times 33 = 132$ cm^3.

51 a)

Volume of C is $5 \times 6 \times 8 = 240$ units^3.

51 b)

Calculator:

$$\begin{vmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{1}{12} \end{vmatrix} = \frac{1}{2}$$

By hand:

$$\begin{aligned} \begin{vmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{1}{12} \end{vmatrix} &= 2\begin{vmatrix} 3 & 0 \\ 0 & \frac{1}{12} \end{vmatrix} - 0\begin{vmatrix} 0 & 0 \\ 0 & \frac{1}{12} \end{vmatrix} + 0\begin{vmatrix} 0 & 3 \\ 0 & 0 \end{vmatrix} \\ &= 2(3 \times \frac{1}{12} - 0 \times 0) - 0 + 0 = 2 \times \frac{1}{4} = \frac{1}{2} \end{aligned}$$

51 c)

Volume of C' is $240 \times \frac{1}{2} = 120$ units^3.

52)

Calculator:

$$\begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ -3 & 2 & 4 \end{vmatrix} = 50$$

By hand:

$$\begin{aligned} \begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ -3 & 2 & 4 \end{vmatrix} &= 2\begin{vmatrix} 0 & 6 \\ 2 & 4 \end{vmatrix} - (-3)\begin{vmatrix} 1 & 6 \\ -3 & 4 \end{vmatrix} + 4\begin{vmatrix} 1 & 0 \\ -3 & 2 \end{vmatrix} \\ &= 2(0 \times 4 - 6 \times 2) - (-3)(1 \times 4 - 6 \times (-3)) + 4(1 \times 2 - 0 \times (-3)) \\ &= 2(-12) + 3 \times 22 + 4 \times 2 = 50 \end{aligned}$$

Volume of S' is $9 \times 50 = 450$ cm^3.

53 a)

Calculator:

$$\begin{vmatrix} 3 & 2 & 0 \\ 1 & 2 & -1 \\ 3 & 3 & -1 \end{vmatrix} = -1$$

By hand:

$$\begin{aligned} \begin{vmatrix} 3 & 2 & 0 \\ 1 & 2 & -1 \\ 3 & 3 & -1 \end{vmatrix} &= 3\begin{vmatrix} 2 & -1 \\ 3 & -1 \end{vmatrix} - 2\begin{vmatrix} 1 & -1 \\ 3 & -1 \end{vmatrix} + 0\begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix} \\ &= 3(2 \times (-1) - (-1) \times 3) - 2(1 \times (-1) - (-1) \times 3) + 0 \\ &= 3 \times 1 - 2 \times 2 = -1 \end{aligned}$$

53 b)

As $\det A = -1$, the transformation preserves volume but reverses orientation.

54 a)

$$\begin{aligned} \begin{vmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & \frac{1}{9} \end{vmatrix} &= 3(6 \times \frac{1}{9} - 0 \times 0) - 0(0 \times \frac{1}{9} - 0 \times 0) + 0(0 \times 0 - 6 \times 0) \\ &= 3 \times 6 \times \frac{1}{9} = 2 \end{aligned}$$

So the volume scale factor is 2.

54 b)

$$\begin{aligned} \begin{vmatrix} -2 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 3 \end{vmatrix} &= -2(18 \times 3 - 0 \times 0) - 0(0 \times 3 - 0 \times 0) + 0(0 \times 0 - 18 \times 0) \\ &= -2 \times 18 \times 3 = -108 \end{aligned}$$

So the volume scale factor is 108.

54 c)

$$\begin{vmatrix} 5 & 1 & 7 \\ 2 & 0 & 0 \\ 5 & 0 & 1 \end{vmatrix} = 5(0 \times 1 - 0 \times 0) - 1(2 \times 1 - 0 \times 5) + 7(2 \times 0 - 0 \times 5) = 0 - 2 + 0 = -2$$

So the volume scale factor is 2.

54 d)

$$\begin{vmatrix} 3 & -2 & -4 \\ 2 & 4 & -3 \\ 3 & 1 & 3 \end{vmatrix} = 3(4 \times 3 - (-3) \times 1) + 2(2 \times 3 - (-3) \times 3) - 4(2 \times 1 - 4 \times 3) = 45 + 30 + 40 = 115$$

So the volume scale factor is 115.

55)

$$\begin{aligned} |A| &= \frac{1}{3}\left(\frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{2}} - 0 \times 0\right) - 0\left(0 \times \frac{1}{\sqrt{2}} - 0 \times 0\right) + 0\left(0 \times 0 - \frac{1}{\sqrt{3}} \times 0\right) \\ &= \frac{1}{3\sqrt{6}} = \frac{\sqrt{6}}{18} \end{aligned}$$

Radius of S is $\sqrt{27} = 3\sqrt{3}$ units.

$$\text{Volume of } S = \frac{4}{3}\pi \times (3\sqrt{3})^3 = 108\sqrt{3}\pi \text{ units}^3$$

$$\text{Volume of } S' = \frac{\sqrt{6}}{18} \times 108\sqrt{3}\pi = 18\sqrt{2}\pi \text{ units}^3$$

56)

Calculator: $\det A = 13$, so $(\det A)^2 = 169$.

$$A^2 = \begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix}\begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} -32 & 31 & 27 \\ -23 & -1 & -28 \\ -17 & -1 & -21 \end{pmatrix}$$

$\det(A^2) = 169$. $\therefore \det(A^2) = (\det A)^2$. By hand:

$$\begin{aligned} \det A &= \begin{vmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{vmatrix} = 2\begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} - 3\begin{vmatrix} -4 & 1 \\ -3 & 1 \end{vmatrix} + 8\begin{vmatrix} -4 & 3 \\ -3 & 2 \end{vmatrix} \\ &= 2(3 \times 1 - 1 \times 2) - 3((-4) \times 1 - 1 \times (-3)) + 8((-4) \times 2 - 3 \times (-3)) \\ &= 2 \times 1 - 3 \times (-1) + 8 \times 1 = 13, \text{ so } (\det A)^2 = 169 \end{aligned}$$

$$A^2 = \begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix}\begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} -32 & 31 & 27 \\ -23 & -1 & -28 \\ -17 & -1 & -21 \end{pmatrix}$$

$$\det(A^2) = \begin{vmatrix} -32 & 31 & 27 \\ -23 & -1 & -28 \\ -17 & -1 & -21 \end{vmatrix}$$

$$= -32\begin{vmatrix} -1 & -28 \\ -1 & -21 \end{vmatrix} - 31\begin{vmatrix} -23 & -28 \\ -17 & -21 \end{vmatrix} + 27\begin{vmatrix} -23 & -1 \\ -17 & -1 \end{vmatrix}$$
$$= -32((-1)(-21) - (-28)(-1)) - 31((-23)(-21) - (-28)(-17))$$
$$+ 27((-23)(-1) - (-1)(-17))$$
$$= -32(-7) - 31(7) + 27(6) = 169$$

$\therefore \det(A^2) = (\det A)^2$.

57)

Calculator: $\det A = -1$, so $(\det A)^3 = -1$.

$$A^3 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 3 & 5 \\ 3 & 4 & 5 \\ 5 & 5 & 7 \end{pmatrix}$$

$\det(A^3) = -1$. $\therefore \det(A^3) = (\det A)^3$. By hand:

$$\det A = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 1\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} - 0\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} + 1\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}$$
$$= 1(1 \times 1 - 1 \times 1) - 0 + 1(0 \times 1 - 1 \times 1) = 1 \times 0 + 1 \times (-1) = -1, \text{ so } (\det A)^3 = -1$$

$$A^3 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 3 & 5 \\ 3 & 4 & 5 \\ 5 & 5 & 7 \end{pmatrix}$$

$$\det(A^3) = \begin{vmatrix} 4 & 3 & 5 \\ 3 & 4 & 5 \\ 5 & 5 & 7 \end{vmatrix} = 4\begin{vmatrix} 4 & 5 \\ 5 & 7 \end{vmatrix} - 3\begin{vmatrix} 3 & 5 \\ 5 & 7 \end{vmatrix} + 5\begin{vmatrix} 3 & 4 \\ 5 & 5 \end{vmatrix}$$
$$= 4(4 \times 7 - 5 \times 5) - 3(3 \times 7 - 5 \times 5) + 5(3 \times 5 - 4 \times 5)$$
$$= 4 \times 3 - 3 \times (-4) + 5 \times (-5) = -1$$

$\therefore \det(A^3) = (\det A)^3$.

58 a)

Calculator:

$$\det A = \begin{vmatrix} 0 & 2 & 0 \\ 3 & 0 & -5 \\ 0 & 4 & 0 \end{vmatrix} = 0$$

By hand:

$$\det A = 0\begin{vmatrix} 0 & -5 \\ 4 & 0 \end{vmatrix} - 2\begin{vmatrix} 3 & -5 \\ 0 & 0 \end{vmatrix} + 0\begin{vmatrix} 3 & 0 \\ 0 & 4 \end{vmatrix}$$
$$= 0 - 2(3 \times 0 - (-5) \times 0) + 0 = -2 \times 0 = 0$$

$\therefore$ the volume of S' is $12 \times 0 = 0$ cm^3.

58 b)

Calculator:

$$\det B = \begin{vmatrix} 5 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 4 \end{vmatrix} = 34$$

By hand:

$$\begin{aligned} \det B &= 5\begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix} - 0\begin{vmatrix} 0 & 0 \\ 3 & 4 \end{vmatrix} + 1\begin{vmatrix} 0 & 2 \\ 3 & 0 \end{vmatrix} \\ &= 5(2 \times 4 - 0 \times 0) - 0 + 1(0 \times 0 - 2 \times 3) = 5 \times 8 + 1 \times (-6) = 34 \end{aligned}$$

$\therefore$ the volume of S' is $12 \times 34 = 408$ cm^3.

58 c)

Calculator:

$$\det C = \begin{vmatrix} -1 & 2 & 1 \\ 3 & -2 & 4 \\ -9 & 1 & 4 \end{vmatrix} = -99$$

By hand:

$$\begin{aligned} \det C &= -1\begin{vmatrix} -2 & 4 \\ 1 & 4 \end{vmatrix} - 2\begin{vmatrix} 3 & 4 \\ -9 & 4 \end{vmatrix} + 1\begin{vmatrix} 3 & -2 \\ -9 & 1 \end{vmatrix} \\ &= -1((-2) \times 4 - 4 \times 1) - 2(3 \times 4 - 4 \times (-9)) + 1(3 \times 1 - (-2) \times (-9)) \\ &= -1 \times (-12) - 2 \times 48 + 1 \times (-15) = -99 \end{aligned}$$

$\therefore$ the volume of S' is $12 \times 99 = 1188$ cm^3.

58 d)

$\det AB = \det A \times \det B = 0 \times 34 = 0$. $\therefore$ the volume of S' is $12 \times 0 = 0$ cm^3.

58 e)

$\det AC = \det A \times \det C = 0 \times -99 = 0$. $\therefore$ the volume of S' is $12 \times 0 = 0$ cm^3.

58 f)

$\det BC = \det B \times \det C = 34 \times -99 = -3366$. $\therefore$ the volume of S' is $12 \times 3366 = 40392$ cm^3.

58 g)

$\det ABC = \det A \times \det B \times \det C = 0 \times 34 \times -99 = 0$. $\therefore$ the volume of S' is $12 \times 0 = 0$ cm^3.

59 a)

$$\begin{vmatrix} k & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & k \end{vmatrix} = k\begin{vmatrix} 2 & 0 \\ 0 & k \end{vmatrix} - 0\begin{vmatrix} 0 & 0 \\ 0 & k \end{vmatrix} + 0\begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix}$$

$= k(2 \times k - 0 \times 0) - 0 + 0 = k \times 2k = 2k^2$.

59 b)

$$\begin{vmatrix} k & 0 & 1 \\ 0 & k & 0 \\ 4 & 0 & k \end{vmatrix} = k \begin{vmatrix} k & 0 \\ 0 & k \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 \\ 4 & k \end{vmatrix} + 1 \begin{vmatrix} 0 & k \\ 4 & 0 \end{vmatrix}$$
$$= k(k \times k - 0 \times 0) - 0 + 1(0 \times 0 - k \times 4) = k \times k^2 + 1 \times (-4k) = k^3 - 4k$$

59 c)

$$\begin{vmatrix} k & 1 & 0 \\ 1 & 2 & 3 \\ 2 & 1 & 4 \end{vmatrix} = k \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} - 1 \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} + 0 \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$$
$$= k(2 \times 4 - 3 \times 1) - 1(1 \times 4 - 3 \times 2) + 0$$
$$= k \times 5 - 1 \times (-2) + 0 = 5k + 2$$

59 d)

$$\begin{vmatrix} k & 0 & k \\ 3 & 5 & -1 \\ 2 & -3 & 2 \end{vmatrix} = k \begin{vmatrix} 5 & -1 \\ -3 & 2 \end{vmatrix} - 0 \begin{vmatrix} 3 & -1 \\ 2 & 2 \end{vmatrix} + k \begin{vmatrix} 3 & 5 \\ 2 & -3 \end{vmatrix}$$
$$= k(5 \times 2 - (-1) \times (-3)) - 0 + k(3 \times (-3) - 5 \times 2)$$
$$= k \times 7 + k \times (-19) = -12k$$

59 e)

$$\begin{vmatrix} 1 & 2 & 4 \\ k & 2 & 1 \\ -4 & 3 & 5 \end{vmatrix} = 1 \begin{vmatrix} 2 & 1 \\ 3 & 5 \end{vmatrix} - 2 \begin{vmatrix} k & 1 \\ -4 & 5 \end{vmatrix} + 4 \begin{vmatrix} k & 2 \\ -4 & 3 \end{vmatrix}$$
$$= 1(2 \times 5 - 1 \times 3) - 2(k \times 5 - 1 \times (-4)) + 4(k \times 3 - 2 \times (-4))$$
$$= 1 \times 7 - 2(5k + 4) + 4(3k + 8) = 2k + 31$$

59 f)

$$\begin{vmatrix} 4 & k & k \\ k & 3 & 2 \\ k & 2 & 3 \end{vmatrix} = 4 \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} - k \begin{vmatrix} k & 2 \\ k & 3 \end{vmatrix} + k \begin{vmatrix} k & 3 \\ k & 2 \end{vmatrix}$$
$$= 4(3 \times 3 - 2 \times 2) - k(k \times 3 - 2 \times k) + k(k \times 2 - 3 \times k)$$
$$= 4 \times 5 - k \times k + k \times (-k) = 20 - 2k^2$$

59 g)

$$\begin{vmatrix} 5 & 3k & 3 \\ -2 & 1 & 4k \\ 2k & 3 & -5 \end{vmatrix} = 5\begin{vmatrix} 1 & 4k \\ 3 & -5 \end{vmatrix} - 3k\begin{vmatrix} -2 & 4k \\ 2k & -5 \end{vmatrix} + 3\begin{vmatrix} -2 & 1 \\ 2k & 3 \end{vmatrix}$$

$$= 5(1 \times (-5) - 4k \times 3) - 3k((-2) \times (-5) - 4k \times 2k) + 3((-2) \times 3 - 1 \times 2k)$$

$$= 5(-5 - 12k) - 3k(10 - 8k^2) + 3(-6 - 2k)$$

$$= -25 - 60k - 30k + 24k^3 - 18 - 6k = 24k^3 - 96k - 43$$

60 a)

$$\begin{vmatrix} k & 1 & 0 \\ 2 & 4 & 3 \\ -1 & 2 & 3 \end{vmatrix} = 1 \Rightarrow k\begin{vmatrix} 4 & 3 \\ 2 & 3 \end{vmatrix} - 1\begin{vmatrix} 2 & 3 \\ -1 & 3 \end{vmatrix} + 0\begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} = 1$$

$$\Rightarrow k(4 \times 3 - 3 \times 2) - 1(2 \times 3 - 3 \times (-1)) + 0 = 1$$

$$\Rightarrow k \times 6 - 1 \times 9 = 1 \Rightarrow 6k - 9 = 1 \Rightarrow k = \frac{5}{3}$$

60 b)

$$\begin{vmatrix} 3 & k & 4 \\ -1 & 2 & -5 \\ 3 & 2 & 7 \end{vmatrix} = 8 \Rightarrow 3\begin{vmatrix} 2 & -5 \\ 2 & 7 \end{vmatrix} - k\begin{vmatrix} -1 & -5 \\ 3 & 7 \end{vmatrix} + 4\begin{vmatrix} -1 & 2 \\ 3 & 2 \end{vmatrix} = 8$$

$$\Rightarrow 3(2 \times 7 - (-5) \times 2) - k((-1) \times 7 - (-5) \times 3) + 4((-1) \times 2 - 2 \times 3) = 8$$

$$\Rightarrow 3 \times 24 - k \times 8 + 4 \times (-8) = 8 \Rightarrow 40 - 8k = 8 \Rightarrow k = 4$$

60 c)

$$\begin{vmatrix} 2 & 3 & -5 \\ -2 & k & 2 \\ 1 & -4 & 1 \end{vmatrix} = -33 \Rightarrow 2\begin{vmatrix} k & 2 \\ -4 & 1 \end{vmatrix} - 3\begin{vmatrix} -2 & 2 \\ 1 & 1 \end{vmatrix} + (-5)\begin{vmatrix} -2 & k \\ 1 & -4 \end{vmatrix} = -33$$

$$\Rightarrow 2(k \times 1 - 2 \times (-4)) - 3(-2 \times 1 - 2 \times 1) - 5(-2 \times (-4) - k \times 1) = -33$$

$$\Rightarrow 2(k + 8) - 3(-4) - 5(8 - k) = -33$$

$$\Rightarrow 2k + 16 + 12 - 40 + 5k = -33 \Rightarrow 7k - 12 = -33 \Rightarrow k = -3$$

60 d)

$$\begin{vmatrix} k & 2 & 5 \\ 9 & 8 & 2 \\ k & 3 & 1 \end{vmatrix} = 49 \Rightarrow k\begin{vmatrix} 8 & 2 \\ 3 & 1 \end{vmatrix} - 2\begin{vmatrix} 9 & 2 \\ k & 1 \end{vmatrix} + 5\begin{vmatrix} 9 & 8 \\ k & 3 \end{vmatrix} = 49$$

$$\Rightarrow k(8 \times 1 - 2 \times 3) - 2(9 \times 1 - 2k) + 5(9 \times 3 - 8k) = 49$$

$$\Rightarrow k \times 2 - 2(9 - 2k) + 5(27 - 8k) = 49$$

$$\Rightarrow 2k - 18 + 4k + 135 - 40k = 49 \Rightarrow 117 - 34k = 49 \Rightarrow k = 2$$

60 e)

$$\begin{vmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k+1 \end{vmatrix} = 150 \Rightarrow k \begin{vmatrix} k & 0 \\ 0 & k+1 \end{vmatrix} - 0 \begin{vmatrix} 0 & 0 \\ 0 & k+1 \end{vmatrix} + 0 \begin{vmatrix} 0 & k \\ 0 & 0 \end{vmatrix} = 150$$

$$\Rightarrow k(k(k+1) - 0 \times 0) - 0 + 0 = 150$$

$$\Rightarrow k^2(k+1) = 150 \Rightarrow k^3 + k^2 - 150 = 0 \Rightarrow k = 5$$

60 f)

$$\begin{vmatrix} 2k & 1 & 3k \\ 1 & 5 & 1 \\ 3k & 1 & 2k \end{vmatrix} = -23 \Rightarrow 2k \begin{vmatrix} 5 & 1 \\ 1 & 2k \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 3k & 2k \end{vmatrix} + 3k \begin{vmatrix} 1 & 5 \\ 3k & 1 \end{vmatrix} = -23$$

$$\Rightarrow 2k(5 \times 2k - 1 \times 1) - (1 \times 2k - 1 \times 3k) + 3k(1 \times 1 - 5 \times 3k) = -23$$

$$\Rightarrow 2k(10k - 1) - (-k) + 3k(1 - 15k) = -23$$

$$\Rightarrow 20k^2 - 2k + k + 3k - 45k^2 = -23$$

$$\Rightarrow 0 = 25k^2 - 2k - 23 = 0 \Rightarrow (25k + 23)(k - 1) = 0$$

$$\Rightarrow k = -\frac{23}{25} \text{ or } k = 1$$

60 g)

$$\begin{vmatrix} k+1 & 2 & 2k \\ 1 & k & 3 \\ k & 0 & k \end{vmatrix} = -2 \Rightarrow (k+1) \begin{vmatrix} k & 3 \\ 0 & k \end{vmatrix} - 2 \begin{vmatrix} 1 & 3 \\ k & k \end{vmatrix} + 2k \begin{vmatrix} 1 & k \\ k & 0 \end{vmatrix} = -2$$

$$\Rightarrow (k+1)(k^2) - 2(k - 3k) + 2k(0 - k^2) = -2$$

$$\Rightarrow (k+1)k^2 - 2(-2k) + 2k(-k^2) = -2$$

$$\Rightarrow k^3 + k^2 + 4k - 2k^3 = -2 \Rightarrow k^3 - k^2 - 4k - 2 = 0$$

$$(k+1)(k^2 - 2k - 2) = 0 \Rightarrow k = -1 \text{ or } k = 1 \pm \sqrt{3}$$

61 a)

When $k = 5$, $A = \begin{pmatrix} 0 & 5 & 2 \\ -23 & 6 & 5 \\ 2 & 1 & \frac{1}{2} \end{pmatrix}$.

$$\det A = 0(6 \times 0 - 5 \times \frac{1}{2}) - 5(-23 \times 0 - 5 \times 2) + 2(-23 \times \frac{1}{2} - 6 \times 2)$$

$$= 0 - 5(-10) + 2(-\frac{47}{2}) = 3$$

$\therefore p = 3$.

61 b)

$$\det A = 3 \Rightarrow \begin{vmatrix} 0 & k & 2 \\ -23 & k+1 & k \\ k-3 & 1 & \frac{1}{2} \end{vmatrix} = 3$$

$$\Rightarrow 0((k+1) \times \frac{1}{2} - k \times 1) - k(-23 \times \frac{1}{2} - k(k-3)) + 2(-23 \times 1 - (k+1)(k-3)) = 3$$

$$\Rightarrow 0 - k(-\frac{23}{2} - k(k-3)) + 2(-23 - (k+1)(k-3)) = 3$$

$$\Rightarrow k^2(k-3) - 23 - 2(k^2 - 2k - 3) = 3$$

$$\Rightarrow k^3 - 3k^2 - 23 - 2k^2 + 4k + 6 - 3 = 0$$

$$\Rightarrow k^3 - 5k^2 + 4k - 20 = 0 \Rightarrow (k-5)(k^2+4) = 0$$

$$\Rightarrow k = 5 \text{ or } k = \pm 2i$$

62)

$$\det A = \begin{vmatrix} k-2 & 2k & 3k+3 \\ 2k-4 & 4k & k+1 \\ 2-k & 8k & 9k+9 \end{vmatrix}$$

$$= (k-2)\begin{vmatrix} 4k & k+1 \\ 8k & 9k+9 \end{vmatrix} - 2k\begin{vmatrix} 2k-4 & k+1 \\ 2-k & 9k+9 \end{vmatrix} + (3k+3)\begin{vmatrix} 2k-4 & 4k \\ 2-k & 8k \end{vmatrix}$$

$$= (k-2)(4k(9k+9) - 8k(k+1))$$
$$- 2k((2k-4)(9k+9) - (k+1)(2-k))$$
$$+ (3k+3)((2k-4)8k - 4k(2-k))$$

$$= (k-2)(36k(k+1) - 8(k+1)k)$$
$$- 2k(18(k-2)(k+1) + (k+1)(k-2))$$
$$+ 3(k+1)(16(k-2)k + 4k(k-2))$$

$$= 36(k-2)(k+1)k - 8(k-2)(k+1)k - 36(k-2)(k+1)k$$
$$- 2(k-2)(k+1)k + 48(k-2)(k+1)k + 12(k-2)(k+1)k$$

$$= 50(k-2)(k+1)k$$

Alternative Method:

$$\begin{vmatrix} k-2 & 2k & 3(k+1) \\ 2(k-2) & 4k & k+1 \\ -(k-2) & 8k & 9(k+1) \end{vmatrix} = (k-2)\begin{vmatrix} 1 & 2k & 3(k+1) \\ 2 & 4k & k+1 \\ -1 & 8k & 9(k+1) \end{vmatrix}$$

$$= 2(k-2)k\begin{vmatrix} 1 & 1 & 3(k+1) \\ 2 & 2 & k+1 \\ -1 & 4 & 9(k+1) \end{vmatrix}$$

$$= 2(k-2)(k+1)k\begin{vmatrix} 1 & 1 & 3 \\ 2 & 2 & 1 \\ -1 & 4 & 9 \end{vmatrix}$$

$$= 2(k-2)(k+1)k \times 25 = 50(k-2)(k+1)k$$

63)

A singular matrix is a square matrix where the determinant is zero (and hence has no inverse).

64 a)

$$\begin{vmatrix} 4 & 2 \\ 4 & 2 \end{vmatrix} = 4 \times 2 - 2 \times 4 = 0$$

So $\begin{pmatrix} 4 & 2 \\ 4 & 2 \end{pmatrix}$ is singular.

64 b)

$$\begin{vmatrix} 3 & 18 \\ 1 & 6 \end{vmatrix} = 3 \times 6 - 18 \times 1 = 0$$

So $\begin{pmatrix} 3 & 18 \\ 1 & 6 \end{pmatrix}$ is singular.

64 c)

$$\begin{vmatrix} -2 & -5 \\ 4 & 10 \end{vmatrix} = -2 \times 10 - (-5) \times 4 = 0$$

So $\begin{pmatrix} -2 & -5 \\ 4 & 10 \end{pmatrix}$ is singular.

64 d)

$$\begin{vmatrix} 6 & -6 \\ -8 & -8 \end{vmatrix} = 6 \times (-8) - (-6) \times (-8) = -96$$

So $\begin{pmatrix} 6 & -6 \\ -8 & -8 \end{pmatrix}$ is not singular.

64 e)

$$\begin{vmatrix} \sqrt{3} & -\sqrt{6} \\ -\sqrt{6} & 2\sqrt{3} \end{vmatrix} = \sqrt{3} \times 2\sqrt{3} - (-\sqrt{6}) \times (-\sqrt{6}) = 0$$

So $\begin{pmatrix} \sqrt{3} & -\sqrt{6} \\ -\sqrt{6} & 2\sqrt{3} \end{pmatrix}$ is singular.

64 f)

$$\begin{aligned} \begin{vmatrix} k+3 & k \\ 2k+6 & 2k \end{vmatrix} &= (k+3) \times 2k - k \times (2k+6) \\ &= 2k^2 + 6k - 2k^2 - 6k = 0 \end{aligned}$$

So $\begin{pmatrix} k+3 & k \\ 2k+6 & 2k \end{pmatrix}$ is singular.

65 a)

$\det A = 4 \times 6 - 3 \times 8 = 0$. $\therefore$ the area of S' is $16 \times 0 = 0$ cm^2.

65 b)

$$\begin{pmatrix} 4 & 3 \\ 8 & 6 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 4a+3b \\ 8a+6b \end{pmatrix}$$

$$y = \frac{8a+6b}{4a+3b}x \Rightarrow y = \frac{2(4a+3b)}{4a+3b}x \Rightarrow y = 2x$$

66 a)

$$\begin{vmatrix} k-3 & 2k+9 \\ 8 & -4 \end{vmatrix} = 0 \Rightarrow (k-3)(-4) - (2k+9)8 = 0$$
$$\Rightarrow -4k + 12 - 16k - 72 = 0 \Rightarrow -20k - 60 = 0 \Rightarrow k = -3$$

66 b)

$$\begin{vmatrix} k-1 & k+5 \\ k+2 & k+3 \end{vmatrix} = 0 \Rightarrow (k-1)(k+3) - (k+5)(k+2) = 0$$
$$\Rightarrow (k^2 + 2k - 3) - (k^2 + 7k + 10) = 0 \Rightarrow -5k - 13 = 0 \Rightarrow k = -\frac{13}{5}$$

66 c)

$$\begin{vmatrix} k+6 & k-4 \\ 3-k & 8-k \end{vmatrix} = 0 \Rightarrow (k+6)(8-k) - (k-4)(3-k) = 0$$
$$\Rightarrow (-k^2 + 2k + 48) - (-k^2 + 7k - 12) = 0 \Rightarrow -5k + 60 = 0 \Rightarrow k = 12$$

66 d)

$$\begin{vmatrix} 2k+3 & k+1 \\ k+2 & k+2 \end{vmatrix} = 0 \Rightarrow (2k+3)(k+2) - (k+1)(k+2) = 0$$
$$\Rightarrow ((2k+3) - (k+1))(k+2) = 0 \Rightarrow (k+2)(k+2) = 0$$
$$\Rightarrow (k+2)^2 = 0 \Rightarrow k = -2$$

66 e)

$$\begin{vmatrix} 3k-2 & k+4 \\ k-2 & k+3 \end{vmatrix} = 0 \Rightarrow (3k-2)(k+3) - (k+4)(k-2) = 0$$
$$\Rightarrow (3k^2 + 7k - 6) - (k^2 + 2k - 8) = 0 \Rightarrow 2k^2 + 5k + 2 = 0$$
$$\Rightarrow (2k+1)(k+2) = 0 \Rightarrow k = -\frac{1}{2} \text{ or } k = -2$$

67)

$$\det A = 3 \times 4 - p \times (-2) = 2p + 12$$
$$\det B = 4 \times q - 2 \times 6 = 4q - 12$$

$$\begin{aligned}\det AB = 0 \Rightarrow \det A \times \det B = 0 \Rightarrow (2p + 12)(4q - 12) = 0 \\ \Rightarrow 8pq - 24p + 48q - 144 = 0 \\ \Rightarrow 8pq + 48q = 24p + 144 \Rightarrow pq + 6q = 3p + 18\end{aligned}$$

68 a)

$\det M = 4 \times \frac{5}{2} - 30 \times \frac{1}{3} = 10 - 10 = 0$. $\therefore$ the matrix M is singular.

68 b)

$\det N = 10 \times (-\frac{1}{3}) - \frac{2}{3} \times (-155) = -\frac{10}{3} + \frac{310}{3} = 100 \neq 0$. $\therefore$ the matrix N is not singular.

68 c)

$\det MN = \det M \times \det N = 0 \times 100 = 0$. $\therefore$ the matrix MN is singular.

68 d)

$\det NM = \det N \times \det M = 100 \times 0 = 0$. $\therefore$ the matrix NM is singular.

69 a)

$$\begin{aligned}\begin{vmatrix}6 & 3 & 4\\ 3 & 0 & 1\\ 2 & -1 & 0\end{vmatrix} &= 6\begin{vmatrix}0 & 1\\ -1 & 0\end{vmatrix} - 3\begin{vmatrix}3 & 1\\ 2 & 0\end{vmatrix} + 4\begin{vmatrix}3 & 0\\ 2 & -1\end{vmatrix} \\ &= 6(0 \times 0 - 1 \times (-1)) - 3(3 \times 0 - 1 \times 2) + 4(3 \times (-1) - 0 \times 2) \\ &= 6 \times 1 - 3 \times (-2) + 4 \times (-3) = 6 + 6 - 12 = 0\end{aligned}$$

So the matrix is singular.

69 b)

$$\begin{aligned}\begin{vmatrix}0 & -6 & 4\\ -1 & 1 & 0\\ 0 & 3 & -2\end{vmatrix} &= 0\begin{vmatrix}1 & 0\\ 3 & -2\end{vmatrix} - (-6)\begin{vmatrix}-1 & 0\\ 0 & -2\end{vmatrix} + 4\begin{vmatrix}-1 & 1\\ 0 & 3\end{vmatrix} \\ &= 0 + 6((-1)(-2) - 0 \times 0) + 4((-1) \times 3 - 1 \times 0) \\ &= 0 \times (-2) + 6 \times 2 + 4 \times (-3) = 0 + 12 - 12 = 0\end{aligned}$$

So the matrix is singular.

69 c)

$$\begin{vmatrix} -1 & 2 & 3 \\ -2 & 3 & -5 \\ 1 & -2 & -2 \end{vmatrix} = (-1)\begin{vmatrix} 3 & -5 \\ -2 & -2 \end{vmatrix} - 2\begin{vmatrix} -2 & -5 \\ 1 & -2 \end{vmatrix} + 3\begin{vmatrix} -2 & 3 \\ 1 & -2 \end{vmatrix}$$
$$= (-1)(3(-2) - (-5)(-2)) - 2((-2)(-2) - (-5)1) + 3((-2)(-2) - 3 \times 1)$$
$$= (-1)(-16) - 2 \times 9 + 3 \times 1 = 16 - 18 + 3 = 1 \neq 0$$

So the matrix is not singular.

69 d)

$$\begin{vmatrix} -\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ -\sqrt{2} & 2\sqrt{8} & 1 \\ \sqrt{2} & -2\sqrt{2} & -\sqrt{8} \end{vmatrix} = \begin{vmatrix} -\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ -\sqrt{2} & 4\sqrt{2} & 1 \\ \sqrt{2} & -2\sqrt{2} & -2\sqrt{2} \end{vmatrix}$$
$$= -\sqrt{2}\begin{vmatrix} 4\sqrt{2} & 1 \\ -2\sqrt{2} & -2\sqrt{2} \end{vmatrix} - 2\sqrt{2}\begin{vmatrix} -\sqrt{2} & 1 \\ \sqrt{2} & -2\sqrt{2} \end{vmatrix} + 2\sqrt{2}\begin{vmatrix} -\sqrt{2} & 4\sqrt{2} \\ \sqrt{2} & -2\sqrt{2} \end{vmatrix}$$
$$= -\sqrt{2}(4\sqrt{2}(-2\sqrt{2}) - 1(-2\sqrt{2})) - 2\sqrt{2}((-\sqrt{2})(-2\sqrt{2}) - 1\sqrt{2})$$
$$+ 2\sqrt{2}((-\sqrt{2})(-2\sqrt{2}) - 4\sqrt{2}\sqrt{2})$$
$$= -\sqrt{2}(-16 + 2\sqrt{2}) - 2\sqrt{2}(4 - \sqrt{2}) + 2\sqrt{2}(-4)$$
$$= 16\sqrt{2} - 4 - 8\sqrt{2} + 4 - 8\sqrt{2} = 0$$

So the matrix is singular.

70 a)

Calculator:

$$\det A = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 5 & 4 & -1 \end{vmatrix} = 6$$

By hand:

$$\det A = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 5 & 4 & -1 \end{vmatrix} = 1\begin{vmatrix} 2 & 1 \\ 4 & -1 \end{vmatrix} - 1\begin{vmatrix} 3 & 1 \\ 5 & -1 \end{vmatrix} + 2\begin{vmatrix} 3 & 2 \\ 5 & 4 \end{vmatrix}$$
$$= 1(2(-1) - 1 \times 4) - 1(3(-1) - 1 \times 5) + 2(3 \times 4 - 2 \times 5)$$
$$= 1(-6) - 1(-8) + 2 \times 2 = 6$$

$\therefore$ the matrix A is not singular.

70 b)

Calculator:

$$\det B = \begin{vmatrix} -2 & 3 & 1 \\ 4 & 1 & 5 \\ 3 & -2 & 1 \end{vmatrix} = 0$$

By hand:

$$\det B = \begin{vmatrix} -2 & 3 & 1 \\ 4 & 1 & 5 \\ 3 & -2 & 1 \end{vmatrix} = -2\begin{vmatrix} 1 & 5 \\ -2 & 1 \end{vmatrix} - 3\begin{vmatrix} 4 & 5 \\ 3 & 1 \end{vmatrix} + 1\begin{vmatrix} 4 & 1 \\ 3 & -2 \end{vmatrix}$$

$$= -2(1 \times 1 - 5(-2)) - 3(4 \times 1 - 5 \times 3) + 1(4(-2) - 1 \times 3)$$
$$= -2 \times 11 - 3(-11) + 1(-11) = 0$$

$\therefore$ the matrix B is singular.

70 c)

$\det AB = \det A \times \det B = 6 \times 0 = 0$. $\therefore$ the matrix AB is singular.

70 d)

$\det BA = \det B \times \det A = 0 \times 6 = 0$. $\therefore$ the matrix BA is singular.

71)

$$\begin{vmatrix} 2 & k & 3 \\ 2 & 2 & k \\ 0 & 1 & -1 \end{vmatrix} = 2\begin{vmatrix} 2 & k \\ 1 & -1 \end{vmatrix} - k\begin{vmatrix} 2 & k \\ 0 & -1 \end{vmatrix} + 3\begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix}$$
$$= 2(2(-1) - k \times 1) - k(2(-1) - k \times 0) + 3(2 \times 1 - 2 \times 0)$$
$$= 2(-2 - k) - k(-2) + 3 \times 2 = -4 - 2k + 2k + 6 = 2 \neq 0$$

So the matrix A is never singular.

72)

$$\begin{vmatrix} k^2 & 1 & 2 \\ -k^2 & 2k^2 & 3 \\ -1 & 1 & -2 \end{vmatrix} = k^2\begin{vmatrix} 2k^2 & 3 \\ 1 & -2 \end{vmatrix} - 1\begin{vmatrix} -k^2 & 3 \\ -1 & -2 \end{vmatrix} + 2\begin{vmatrix} -k^2 & 2k^2 \\ -1 & 1 \end{vmatrix}$$
$$= k^2(2k^2(-2) - 3 \times 1) - 1((-k^2)(-2) - 3(-1)) + 2((-k^2)1 - 2k^2(-1))$$
$$= k^2(-4k^2 - 3) - (2k^2 + 3) + 2k^2 = -4k^4 - 3k^2 - 2k^2 - 3 + 2k^2$$
$$= -4k^4 - 3k^2 - 3$$

$k^4 \geq 0$ for all real values of k, so $-4k^4 \leq 0$ for all real values of k. $k^2 \geq 0$ for all real values of k, so $-3k^2 \leq 0$ for all real values of k. $\therefore -4k^4 - 3k^2 - 3 < 0$ for all real values of k, and so can never be equal to zero. So the matrix A is never singular.

73)

$$\det A = -\sin\theta\begin{vmatrix} 1 & 0 \\ 1 & \sin\theta \end{vmatrix} - 1\begin{vmatrix} 0 & 0 \\ \cos\theta & \sin\theta \end{vmatrix} + \cos\theta\begin{vmatrix} 0 & 1 \\ \cos\theta & 1 \end{vmatrix}$$
$$= -\sin\theta(1 \times \sin\theta - 0 \times 1) - 1(0 \times \sin\theta - 0 \times \cos\theta)$$
$$+ \cos\theta(0 \times 1 - 1 \times \cos\theta)$$
$$= -\sin\theta \times \sin\theta - 1 \times 0 + \cos\theta \times (-\cos\theta)$$
$$= -\sin^2\theta - \cos^2\theta = -(\sin^2\theta + \cos^2\theta) = -1 \neq 0$$

So the matrix A is never singular.

74)

$$\begin{aligned}
\det A &= (\sin\theta - 1)\begin{vmatrix} 1 & 0 \\ 1 & 1-\sin\theta \end{vmatrix} - 1\begin{vmatrix} 0 & 0 \\ 1-\cos\theta & 1-\sin\theta \end{vmatrix} \\
&\quad + (1-\cos\theta)\begin{vmatrix} 0 & 1 \\ 1-\cos\theta & 1 \end{vmatrix} \\
&= (\sin\theta - 1)(1\times(1-\sin\theta) - 0\times 1) \\
&\quad - 1(0\times(1-\sin\theta) - 0\times(1-\cos\theta)) \\
&\quad + (1-\cos\theta)(0\times 1 - 1\times(1-\cos\theta)) \\
&= (\sin\theta - 1)(1-\sin\theta) + (1-\cos\theta)(\cos\theta - 1) \\
&= 2\sin\theta - \sin^2\theta - 1 + 2\cos\theta - 1 - \cos^2\theta \\
&= -(\sin^2\theta + \cos^2\theta) + 2\sin\theta + 2\cos\theta - 2 \\
&= -1 + 2\sin\theta + 2\cos\theta - 2 \\
&= 2\sin\theta + 2\cos\theta - 3 \\
&= 2\sqrt{2}\sin(\theta + \frac{\pi}{4}) - 3
\end{aligned}$$

The maximum value of $2\sqrt{2}\sin(\theta+\frac{\pi}{4}) - 3$ is $2\sqrt{2} - 3 = -0.17157\ldots < 0$. So the matrix A is never singular.

75 a)

$$\begin{vmatrix} 7 & -1 & k \\ 2 & 1 & k \\ 0 & 5 & 4 \end{vmatrix} = 0 \Rightarrow 7\begin{vmatrix} 1 & k \\ 5 & 4 \end{vmatrix} - (-1)\begin{vmatrix} 2 & k \\ 0 & 4 \end{vmatrix} + k\begin{vmatrix} 2 & 1 \\ 0 & 5 \end{vmatrix} = 0$$

$$\Rightarrow 7(1\times 4 - k\times 5) - (-1)(2\times 4 - k\times 0) + k(2\times 5 - 1\times 0) = 0$$
$$\Rightarrow 7(4-5k) + 1\times 8 + k\times 10 = 0$$
$$\Rightarrow 28 - 35k + 8 + 10k = 0 \Rightarrow 36 - 25k = 0 \Rightarrow k = \frac{36}{25}$$

75 b)

$$\begin{vmatrix} -2 & k & 4 \\ -3 & 1 & k \\ 1 & 2 & 4 \end{vmatrix} = 0 \Rightarrow -2\begin{vmatrix} 1 & k \\ 2 & 4 \end{vmatrix} - k\begin{vmatrix} -3 & k \\ 1 & 4 \end{vmatrix} + 4\begin{vmatrix} -3 & 1 \\ 1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow -2(1\times 4 - k\times 2) - k((-3)\times 4 - k\times 1) + 4((-3)\times 2 - 1\times 1) = 0$$
$$\Rightarrow -2(4-2k) - k(-12-k) + 4(-7) = 0$$
$$\Rightarrow -8 + 4k + 12k + k^2 - 28 = 0 \Rightarrow k^2 + 16k - 36 = 0$$
$$\Rightarrow (k+18)(k-2) = 0 \Rightarrow k = -18 \text{ or } k = 2$$

75 c)

$$\begin{vmatrix} k+2 & 2 & -3 \\ 2 & -3 & 2 \\ 1 & 0 & k-2 \end{vmatrix} = 0 \Rightarrow (k+2)\begin{vmatrix} -3 & 2 \\ 0 & k-2 \end{vmatrix} - 2\begin{vmatrix} 2 & 2 \\ 1 & k-2 \end{vmatrix} + (-3)\begin{vmatrix} 2 & -3 \\ 1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (k+2)(-3(k-2)-2\times 0)-2(2(k-2)-2\times 1)$$
$$+(-3)(2\times 0-(-3)\times 1)=0$$
$$\Rightarrow (k+2)(6-3k)-2(2k-6)-3\times 3=0$$
$$\Rightarrow -3k^2+12-4k+12-9=0 \Rightarrow 0=3k^2+4k-15$$
$$\Rightarrow 0=(3k-5)(k+3) \Rightarrow k=\frac{5}{3} \text{ or } k=-3$$

76)

A', B', C', D' and E' are coplanar if they lie in the same plane, so the image S' will have a volume of 0 cm^3. This will occur if $\det A = 0$.

$$\begin{vmatrix} 2k & 1 & k \\ 4k & 5 & 3 \\ 3 & -6 & k \end{vmatrix} = 0 \Rightarrow 2k\begin{vmatrix} 5 & 3 \\ -6 & k \end{vmatrix} - 1\begin{vmatrix} 4k & 3 \\ 3 & k \end{vmatrix} + k\begin{vmatrix} 4k & 5 \\ 3 & -6 \end{vmatrix} = 0$$
$$\Rightarrow 2k(5k-3(-6))-(4k^2-3\times 3)+k(4k(-6)-5\times 3)=0$$
$$\Rightarrow 2k(5k+18)-(4k^2-9)+k(-24k-15)=0$$
$$\Rightarrow 10k^2+36k-4k^2+9-24k^2-15k=0$$
$$\Rightarrow 0=18k^2-21k-9 \Rightarrow 0=6k^2-7k-3$$
$$\Rightarrow 0=(3k+1)(2k-3) \Rightarrow k=-\frac{1}{3} \text{ or } k=\frac{3}{2}$$

77)

Let A and B be two square matrices.

$$\text{If } \det A=0, \text{ then } \det AB=\det A\times\det B=0\times\det B=0$$

$$\text{If } \det B=0, \text{ then } \det AB=\det A\times\det B=\det A\times 0=0$$

$$\text{If } \det A=0 \text{ and } \det B=0, \text{ then } \det AB=\det A\times\det B=0\times 0=0$$

$\therefore$ if at least one of two square matrices are singular, then their product will be a singular matrix.

78)

Let $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $\det A=ad-bc\neq 0$. Let $B=\begin{pmatrix} p & q \\ r & s \end{pmatrix}$, then $\det B=ps-qr\neq 0$.

$$\det A+\det B=ad-bc+ps-qr$$

$$A+B=\begin{pmatrix} a & b \\ c & d \end{pmatrix}+\begin{pmatrix} p & q \\ r & s \end{pmatrix}=\begin{pmatrix} a+p & b+q \\ c+r & d+s \end{pmatrix}$$

$$\begin{aligned}\det(A+B)&=(a+p)(d+s)-(b+q)(c+r)\\&=ad+as+pd+ps-bc-br-qc-qr\\&=(ad-bc+ps-qr)+(as+pd-br-qc)\end{aligned}$$

$$ad-bc+ps-qr\neq(ad-bc+ps-qr)+(as+pd-br-qc)$$

$\therefore \det(A+B)\neq\det A+\det B$.

79)

Let $A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$, then $\det A = 1 \neq 0$. Let $B = \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$, then $\det B = -1 \neq 0$.

$$\det A + \det B = 1 + (-1) = 0$$

$$A + B = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 0 \end{pmatrix}$$

$$\det(A + B) = 2 \times 0 - 2 \times 0 = 0$$

$\therefore \det(A + B) = \det A + \det B$ in this case. Alternative Method: Consider A and B as rotations about the origin by 45°, with one being anticlockwise and the other being clockwise.

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \text{ then } \det A = 1 \neq 0$$

$$B = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \text{ then } \det B = 1 \neq 0$$

$\det A + \det B = 1 + 1 = 2$.

$$A + B = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} + \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{2}} & 0 \\ 0 & \frac{2}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$$

$\det(A + B) = \sqrt{2} \times \sqrt{2} - 0 \times 0 = 2$. $\therefore \det(A + B) = \det A + \det B$ in this case.

80)

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $\det A = ad - bc \neq 0$. Let $B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, then $\det B = ps - qr \neq 0$.

$$\det A \times \det B = (ad - bc)(ps - qr) = adps - adqr - bcps + bcqr$$

$$AB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}$$

$$\begin{aligned} \det AB &= (ap + br)(cq + ds) - (aq + bs)(cp + dr) \\ &= apcq + apds + brcq + brds - aqcp - aqdr - bscp - bsdr \\ &= acpq + adps + bcqr + bdrs - acpq - adqr - bcps - bdrs \\ &= adps + bcqr - adqr - bcps \\ &= adps - adqr - bcps + bcqr \end{aligned}$$

$\therefore \det AB = \det A \times \det B$.